

Hspice Simulation

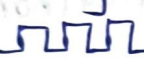
01 Noise [crosstalk]

1.1 what is crosstalk in TSV model?

01 It is a noise effect.

02 unwanted electrical interference where one TSV (aggressor) $\uparrow$ freq signal leak to other TSV (victim) through Si Sub.

1.2 Aggressor: The TSV which is taking input (switching 0 to 1 too-)

 $\rightarrow$  E field & M field.

1.3 Victim: Nearby TSV to aggressor that is supposed to be quiet [connected to GND] but accidentally picks up the EM field of aggressor.

1.4 Crosstalk

1.4.1 Crosstalk Glitch

The spike that occurs in a steady wave of victim TSV due to the switching of aggressor TSV.

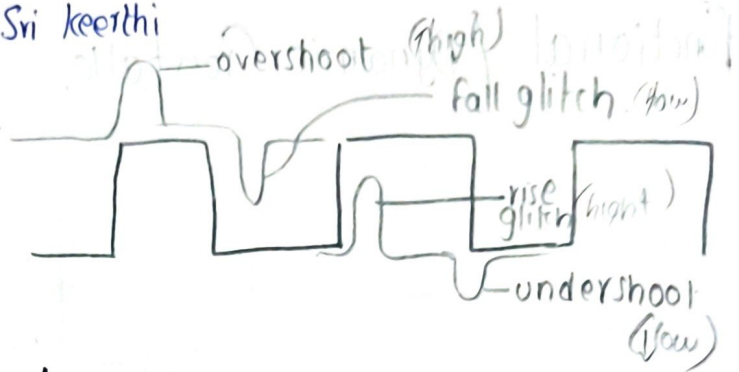
(sudden 0 to 1 q is pumped the coupling C effects voltage spikes)

Rise Glitch / Fall Glitch

Ideal case



Crosstalk Glitch case



1.4.2 Crosstalk delay

01 Both the aggressor and victim getting input at same time.

02 Both switching charging & discharging behavior of coupling capacitance changes drastically due to this the transition time of victims speeds up or slows down.

Time take for digital signal to switch from one logic state 0 to 1 or 1 to 0.

+ve crosstalk delay: (slows down)



victim have to work hard twice.
delay time $\uparrow$

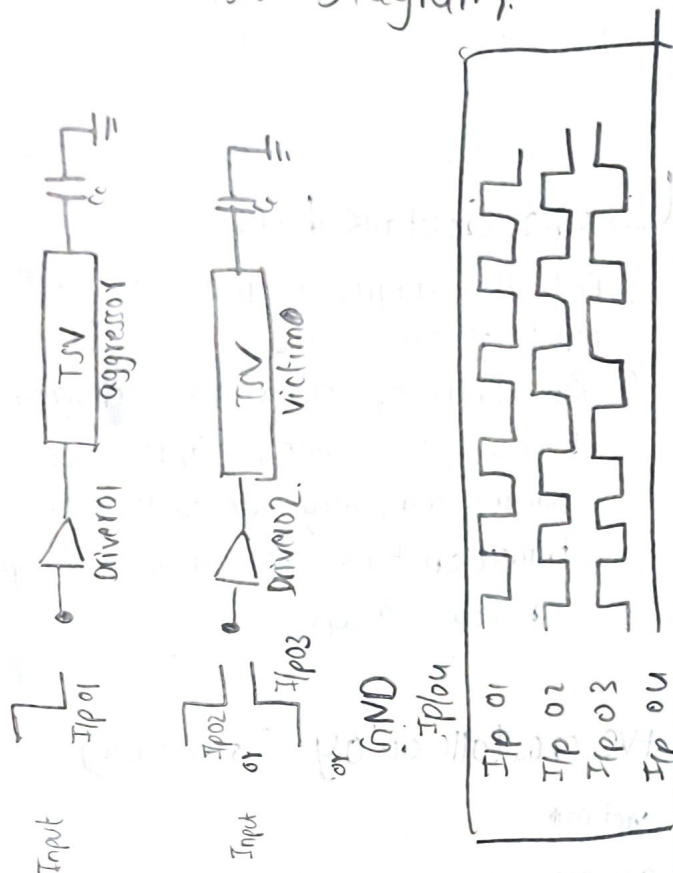
+ve crosstalk delay (speedup)



not hard and even aggressor essentially helps the Cap for victim to switch faster than normal along with it.

functional / Dynamic Crosstalk.

Driver - TSV Diagram.



functional crosstalk.

- aggressor takes I/p01
- victim takes I/p04

Dynamic crosstalk.

01 In phase

- Agg. takes I/p01
- victim takes I/p02

02 Out phase

- Agg. takes I/p01
- victim takes I/p03.

Miller Coupling Capacitor.

$$C_{eff} = C_c \times (1 - A_v)$$

C_{eff} = effective miller capacitance
~~seen by~~

C_c = Coupling Capacitance btw two TSVs

A_v = Voltage gain or aggressor voltage change to victim's voltage change

$$\left[\frac{\Delta V_{aggressor}}{\Delta V_{victim}} \right]$$

Meaning: what form of capacitance does victim see (form of coupling capacitance what form?)

So there is a coupling sitting btw aggressor and victim so in what form that victim see this coupling capacitance.

→ functional. $(\Delta V_{aggressor} = 0, A_v = 0)$
 $C_{eff} = C_c$

→ In phase.
 $C_{eff} = 0$

$$\Delta V_{aggressor} = A_v \Delta V_{victim} \quad A_v = 1$$

→ out phase.
 $C_{eff} = 2C_c$

$$\Delta V_{aggressor} = -\Delta V_{victim}$$

- crosstalk Simulation Defect free.

→ Refer. Day 20 1st model for defect-free for both circuit and Network

Code: *QA*

- temp 25

**** Supply ****

Vagg 1 0 pulse (0 1.8 0s 100p 100p 5n 10n)

Vg01 5 0 DC 0

Vg02 11 0 DC 0

*** Network ***

(Refer Day 20 1st model)

*** Operation ***

- Option post
- tran 1p 100n

*** measure**

- PROBE V(1) V(3) V(7) V(9) V(10)
- end.